

LAMPHUN, Thailand

WMO: 483290

Lat: 18.5668N

Lon: 99.0384E

Elev: 298

StdP: 97.79

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	12.4	13.9	8.0	6.9	25.2	9.6	7.7	24.0	3.4	25.5	3.0	25.7	0.0	60	0.289

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	13.0	38.8	23.5	37.6	23.3	36.4	23.2	26.7	32.2	26.4	31.7	26.1	31.3	1.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.5	21.5	28.3	25.2	21.1	27.9	24.9	20.7	27.7	85.8	32.4	84.4	31.9	83.1	31.5	31.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
(a)	(b)	(c)		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
4.7	3.6	3.0	DB	10.0	40.5	1.8	1.2	8.7	41.3	7.7	42.0	6.7	42.7	5.4	43.6	
			WB	9.3	28.2	1.7	1.2	8.1	29.1	7.1	29.8	6.2	30.5	5.0	31.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	22.8	25.0	28.1	30.7	29.7	29.0	28.5	28.0	27.8	27.0	25.1	22.8
	DBStd	3.03	1.89	1.91	1.84	1.96	2.17	1.32	1.29	1.15	1.08	1.42	1.97	2.40
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	3	1	0	0	0	0	0	0	0	0	0	0	2
	CDD10.0	6221	398	420	562	621	609	571	573	559	533	526	452	396
	CDD18.3	3183	141	187	304	371	351	321	315	301	283	267	202	140
	CDH23.3	33820	1230	2034	3747	4924	4155	3646	3416	3033	2677	2301	1590	1067
	CDH26.7	15217	446	1010	2163	2917	2077	1541	1332	1081	933	813	573	333
(14) Wind	WSAvg	0.5	0.2	0.4	0.6	0.7	0.7	0.8	0.7	0.5	0.3	0.2	0.2	0.2
(15) Precipitation	PrecAvg	1091	9	8	22	52	167	119	139	198	216	108	41	11
	PrecMax	1641	70	43	129	201	329	245	242	449	422	209	224	83
	PrecMin	729	0	0	0	7	12	23	69	84	105	25	0	0
	PrecStd	211	16	12	28	44	78	53	52	78	69	43	45	19
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	36.4	39.1	40.7	39.6	36.6	35.2	34.3	34.0	33.7	33.5	32.4
		MCWB	21.1	20.8	22.1	23.8	24.6	25.6	25.4	25.5	25.9	25.6	24.1	23.2
	2%	DB	31.7	34.9	37.7	39.5	37.7	34.8	33.9	33.2	33.0	32.7	32.2	31.1
		MCWB	20.9	20.7	21.8	23.6	24.8	25.5	25.3	25.4	25.8	25.3	24.0	22.5
	5%	DB	30.6	33.7	36.6	38.3	36.1	33.7	33.0	32.3	32.1	31.8	31.2	30.0
		MCWB	20.7	20.6	21.7	23.6	24.9	25.4	25.2	25.3	25.6	25.0	23.7	22.0
	10%	DB	29.4	32.4	35.4	37.0	34.5	32.6	32.0	31.4	31.2	30.8	30.0	28.7
		MCWB	20.4	20.5	21.6	23.6	25.0	25.3	25.1	25.2	25.4	24.9	23.2	21.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.1	24.6	26.3	26.6	26.9	26.9	26.8	26.7	26.9	26.6	25.7	25.1
		MCDB	29.8	32.8	35.1	36.3	33.7	32.8	31.9	31.4	31.6	31.5	30.8	30.1
	2%	WB	22.9	23.3	24.4	25.6	26.3	26.5	26.2	26.2	26.5	26.1	25.1	23.8
		MCDB	28.6	31.0	33.0	34.9	33.0	32.0	31.2	30.8	30.9	30.8	29.9	28.8
	5%	WB	22.1	22.4	23.6	25.1	26.0	26.1	25.9	25.9	26.1	25.7	24.6	22.9
		MCDB	27.5	29.8	32.0	34.0	32.6	31.3	30.6	30.3	30.4	30.0	29.0	27.8
	10%	WB	21.4	21.6	22.9	24.7	25.6	25.8	25.5	25.6	25.7	25.2	24.0	22.2
		MCDB	27.0	29.2	31.3	33.0	31.7	30.6	30.0	29.6	29.6	29.2	28.2	26.9
(35) Mean Daily Temperature Range	5% DB	MDBR	14.8	16.6	15.7	13.0	9.2	7.5	7.1	7.0	7.3	8.1	10.6	12.6
		MCDBR	16.4	18.1	17.6	14.8	11.6	9.0	8.6	8.4	8.6	9.5	11.6	13.4
		MCWBR	6.3	5.9	4.9	3.3	2.6	2.3	2.3	2.3	2.5	2.9	4.1	5.2
	5% WB	MCDBR	13.0	15.1	14.3	12.5	9.5	7.9	7.6	7.4	7.8	8.5	9.8	11.5
		MCWBR	5.6	5.9	4.9	3.4	2.7	2.4	2.4	2.4	2.6	2.9	3.7	4.7
(40) Clear-Sky Solar Irradiance	taub		0.436	0.513	0.671	0.667	0.504	0.439	0.427	0.442	0.475	0.517	0.443	0.437
	taud		2.199	1.995	1.637	1.708	2.172	2.386	2.418	2.359	2.248	2.085	2.258	2.223
	Ebn at Noon		838	791	684	692	806	852	863	857	829	778	827	824
	Edh at Noon		137	176	261	245	152	122	118	126	139	159	128	130
(44) All-Sky Solar Radiation	RadAvg		4.60	5.37	5.65	6.02	5.37	4.67	4.34	4.30	4.54	4.45	4.59	4.32
	RadStd		0.23	0.19	0.37	0.35	0.46	0.33	0.33	0.27	0.20	0.25	0.32	0.25

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	+0.19	+0.23	N/A	+1	N/A	N/A
Regional (17 neighbors)	+0.27	N/A	N/A	N/A	N/A	+0.15	N/A	N/A	+138	+139

Nomenclature: See separate page